Vincent T. Cook

MANUFACTURING & PRODUCTION

San Jose, CA | (805) 397-9594 | Vincent.Tait.Cook@gmail.com
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Aerospace engineering senior with CAD assemblies, schematics, motor test-bench documentation, and electrical troubleshooting experience. Background includes structural lab testing, wind-tunnel measurements, and mechanical project planning.

EDUCATION

San Jose State University | San Jose, CA
B.S. Aerospace Engineering | Expected May 2027 | Enrolled Sep 2022

TECHNICAL SKILLS

Design: SolidWorks, Autodesk Inventor, 3D CAD, schematics, assembly instructions, stress analysis
Testing: circuit repairs, battery testing, MATLAB/Simulink, Python, CFD, thermal testing, simulation

ENGINEERING EXPERIENCE & PROJECTS

SAE Spartan Racing | EV Powertrain Intern | Aug 2023-May 2025
- Developed CAD and assembly instructions for an electric motor test bench; integrated mechanical/electrical components and aligned engineering/manufacturing practices.

Hybrid Electric / Hydrogen Fuel-Cell Aircraft | SJSU Capstone
Aerospace Engineering Team Member - In Progress | 2026-Present
- Contribute wing-model designs and project planning. Planned work includes CAD, CFD/thermal analysis, fabrication/testing, and fuel/battery electronics; team flight target spring 2027.

SJSU Robotics | Mechanical Designer | Aug 2024-Present
- Designed a transmitter mast with camera/sensor mounts; used stress analysis to reinforce high-load regions and select lighter materials elsewhere.

BROADSWORD | Senior Design Rocket Concept | Mechanical & Programming Subgroup Lead | Aug-Dec 2023
- Designed reusable rocket landing legs in Inventor and coded stage liftoff/landing trajectories; coordinated mechanical/programming tasks for a student concept project.

NASA L'SPACE Mission Concept Academy
Mechanical Engineering Team Member - Educational Program | Fall 2026-Present
- Help plan science approaches and evaluate mechanical components for a robotic lunar payload mission concept.

FIRST Robotics Team 3925 | Electrical Lead - High School Robotics | 2020-2022
- Led robot circuitry design, battery testing, and circuit repairs; developed programming knowledge alongside electrical work.

SELECTED COURSEWORK

- AE 20 / ENGR 10 | Modeled a V6 piston engine in Inventor with paper schematics; completed wind-tunnel blade/structural design, power-efficiency studies, and robot design/build/programming.
- AE 112 / AE 114 | Tested structural/building designs and compared calculations in labs; studied material properties.
- AE 160 / AE 162 | Tested objects in an open wind tunnel; analyzed lift, drag, pitch, coefficients, and vortices; studied Navier-Stokes equations in aerodynamics coursework.

WORK EXPERIENCE

PayPal Stadium | Guest Representative | 2026 Soccer Season
- Worked San Jose Earthquakes and Bay FC women's team matches.

The Tech Interactive | Educator I | 2024-2025
- Led STEM labs/design challenges and explained procedures and technical concepts.

Self-Employed | Private Swim Instructor | 2023
- Scheduled clients and tailored swim lessons to individual needs.

YMCA & Ventura Aquatics Center | Swim Instructor / Coach / Lifeguard | 2021-2024
- Balanced coaching/lifeguard work and adapted swim lessons to varied abilities.

LEADERSHIP & ATHLETICS

SJSU Division I Club Ultimate | 2022-Present: coordinate gym training and conditioning. SJSU Club Swimming | Spring 2026: competed at a Cal-hosted meet.